

Exponential Expressions

Teacher's Edition

The Period

MA.7.NSO.1.1 Know and apply the laws of exponents to evaluate numerical expressions and generate equivalent numerical expressions, limited to whole-number exponents and rational-number bases.

Must Two or three laws in one expression, whole-number exponents and rational bases; generate an equivalent expression and evaluate it.

Must not No negative exponents yet. The laws never cross addition or subtraction, and order of operations stays at six steps or fewer.

Target I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

MTR.2.1 — Notes II and board 9 · **MTR.3.1** — Boards 1 through 4 ·

MTR.6.1 — Board 5

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up	2-3	5 min
Notes	4-6	10 min
Example 1	7-8	3 min
Your Turn	9-10	3 min
Example 2	11-12	3 min
Your Turn	13-14	3 min
Whiteboards	15-32	14 min
Independent Set	33	6 min
IXL	34	5 min
Total		53 min

Say these three things out loud today

1. Make the plan before you make a mark. Inside the parentheses first, or expand first — either, but pick one.
2. A zero exponent is a 1 sitting in the expression. It changes nothing, and it is not nothing.
3. The laws work on multiplication and division only. They never cross a plus sign.

Slide by Slide

1. **Exponential Expressions.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q4 is the prerequisite: exponents before multiplication before addition.
3. **Warm-Up — Answers.** Reveal. 1. 100 2. 7 3. $x = 5$ 4. 50
4. **Notes — Copy the table. It is your reference for the rest of the unit.** This table is the durable record: students have no worksheet and no study guide, so the notes are what they keep (ruling 11).
5. **Notes — Two students, two orders, one value.** Pick the order that makes the numbers smallest soonest. Inside the parentheses first is usually that order.
6. **Notes.** The laws are shortcuts for multiplying and dividing powers.
7. **Example 1 — A zero exponent inside a product inside a quotient.** SPOKEN ONLY.
8. **Example 1 — Worked — Caroline's plan and Jamir's plan.** Caroline adds $5 + 0$ first (product of powers), then subtracts.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a student who reads 5^0 as 0 (the whole product becomes 0) and one who adds $4 + 0 + 2$ (5^6).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Three laws, a decimal and a negative base.** SPOKEN ONLY.
12. **Example 2 — Worked.** Zero exponent, then power of a quotient backwards, then one power of -10 .
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches a dropped sign (27) and a student who applied the zero law to the wrong power ($7 \cdot$ something).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.** A warehouse holds 6^4 boxes in 6^2 equal rows. Each box weighs 3 kilograms. How many boxes are in each row? → **36** *The weight of 3 kilograms is not needed. 1296 — the 6^2 rows were ignored [the quotient was never taken]; 6^6 or 46,656 — the exponents were added [confused with product of powers]; anything built from 3 — the unneeded figure was used*
- 17-18. **Board 2.** $(3^0 \cdot 3^2)^3$ → **729** *243 — added the exponents $2 + 3$ in the power of a power step [confused with product of powers]; 0 — zero exponent read as 0*
- 19-20. **Board 3.** $\frac{(0.5^3)^2}{(0.5)^2}$ → **0.0625** *0.125 — divided the exponents $6 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]; 0.015625 — 0.5^6 , the denominator ignored*
- 21-22. **Board 4.** $\left(\frac{2.5^5}{2.5^3}\right)^1$ → **6.25** *2.5 — the outside 1 was treated as the final exponent [identity law misapplied]; 15.625 — used 2.5^3*
- 23-24. **Board 5.** $\frac{(2^2 \cdot 3.2^0)^6}{2^{10}}$ Which is the value? → **D — 4** *(A) $(2^2)^6$ written as 2^8 by adding $2 + 6$, then $2^8 \div 2^{10}$ [power of a power confused with product of powers]; (B) added the exponents $12 + 10$ instead of subtracting [confused with product of powers]; (C) 3.2^0*

read as 0, so the whole numerator is 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

25-26. Board 6. $\frac{0.2^3}{2^3} \cdot 0.1^2$ Which expression is equivalent? → **C — 0.1^5** (A) quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]; (B) the 2^3 in the denominator was ignored [order of operations: the fraction bar is a grouping]; (D) multiplied the exponents $3 \cdot 2$ [confused with power of a power] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

27-28. Board 7. $0.1^2 \cdot (0.1^1)^2$ → **0.0001** $0.001 = 0.1^3$, the outer square on $(0.1^1)^2$ dropped, leaving $0.1^2 \cdot 0.1^1$ [power of a power step skipped]; $0.01 = 0.1^2$

29-30. Board 8. $\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$ → **-64** 64 — dropped the sign; 'undefined' — $(-4)^0$ in the denominator read as 0, so a division by 0 [8.NSO.1.3: zero exponent law, $a^0 = 1$]

31-32. Board 9. Winston writes $(4^7/4^4)^2 = (4^3)^2$. Yvette writes $(4^7/4^4)^2 = 4^{14}/4^8$. Do they get the same value? Explain. → **Yes — both give $4^6 = 4,096$** 'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

33. Independent Set — Six questions. On your own, in writing. Hand out the Independent Set; the same six questions are on the board.

34. IXL — Last five minutes. IXL: every listed skill is required, SmartScore 67.

Misconceptions to Watch

Board 1. 1296 — the 6^2 rows were ignored [the quotient was never taken]

Board 2. 243 — added the exponents $2 + 3$ in the power of a power step [confused with product of powers]

Board 3. 0.125 — divided the exponents $6 \div 2$ [7.NSO.1.1 misconception: 'divide' applied to the exponents]

Board 4. 2.5 — the outside 1 was treated as the final exponent [identity law misapplied]

Board 5. $(2^2)^6$ written as 2^8 by adding $2 + 6$, then $2^8 \div 2^{10}$ [power of a power confused with product of powers]

Board 6. quotient of powers applied to DIFFERENT bases, 0.2 and 2 [7.NSO.1.1: the quotient law needs the same base]

Board 7. $0.001 = 0.1^3$, the outer square on $(0.1^1)^2$ dropped, leaving $0.1^2 \cdot 0.1^1$ [power of a power step skipped]

Board 8. 64 — dropped the sign

Board 9. 'No' — judged equivalence by appearance instead of value [7.NSO.1.1: generate equivalent expressions]

Materials Whiteboards and markers. Calculators are allowed on everything; the decimal values in the bank (5.0625, 2,293.45007, 9,605.9601) are calculator work once the single power is found.